

Alex Lu Chen

aluchen@stanford.edu | alexluchen.github.io

EDUCATION

Stanford University

M.S. in Computer Science (incoming)

Stanford, CA

Expected June 2028

University of California, Los Angeles (UCLA)

B.S. in Computer Science

Los Angeles, CA

GPA 3.97/4.00, June 2026

Relevant courses: Machine Learning, Natural Language Processing, Computer Vision, Data Structures & Algorithms, Software Construction

- Awards and Honors: Tau Beta Pi, Upsilon Pi Epsilon

WORK EXPERIENCE

Ramp

Applied AI (Intern)

June 2026 – Aug 2026

San Francisco, CA

- Implementing embedding-based similarity matching and heuristics to reevaluate AI-assisted transaction categorizations after manual overrides

Amazon

Software Engineer (Intern)

Jan 2025 – April 2025

Seattle, Washington

- Designed and deployed a full-stack internal tool to trace which ML-based strategies led to specific Amazon.com recommendations, enabling large-scale analysis across millions of product traces
- Retrieved and transformed S3-stored model outputs using AWS Athena (SQL) to match key-value data with customer history; implemented with Java AWS Lambda, TypeScript CDK, and React frontend, deployed via AWS CodePipeline (CI/CD)

Physics and Astronomy Department, UCLA

Software Engineer (Part-Time)

Nov 2023 – Nov 2024

Los Angeles, CA

- Spearheaded the development of Bruin Bot, a ChatGPT-based MERN-stack chatbot, enhancing academic resources for the UCLA community
- Deployed Bruin Bot using Nginx, used by 50 research lab members and securing \$35,000 in funding from the Physics department

RESEARCH & PROJECTS

MINT Lab, UCLA

Research Assistant, Advisor: Aditya Grover

June 2025 – Present

Los Angeles, CA

- First author of *The Pitfalls of KV Cache Compression* (ACL main conference), revealing overlooked efficiency-accuracy trade-offs in LLM inference; project website: pitfalls-of-kv-cache-compression.github.io
- Implemented the majority of the PyTorch codebase, enabling reproducible evaluations for various KV-cache compression methods across multi-instruction LLM benchmarks

Text-to-Speech Model in PyTorch

Feb 2025 – June 2025

Remote

- Built a custom text-to-speech model from scratch in PyTorch, using phoneme-based text tokenization, Descript Audio Codec for audio compression, and an autoregressive transformer to generate audio token sequences
- Achieved functional speech generation within 5 hours of training on 4× NVIDIA T4 GPUs
- Shared a detailed write-up at alexluchen.github.io/AlexTTS with full source code at github.com/alexluchen/AlexTTS

SKILLS

Languages:

Python, Java, C++, TypeScript, SQL

Systems & Tools:

Git, Linux, AWS, Datadog, Temporal

ML & Data:

PyTorch, NumPy, CUDA, Snowflake